Friedrich-Alexander-Universität
Erlangen-Nürnberg
Naturwissenschaftliche Fakultät

Lehrstuhl
Mathematik und
Data Science

MASTER THESIS

Hybrid AI Framework for Employee Project Allocation (HAF-EPA)

submitted in partial fulfillment of the requirements for the degree of

Master of Data Science

Submitted by

MD Firozur Rahman

Matriculation Number: **22975954**

First Supervisor
Prof. Frauke Liers

Second Supervisor
Prof. Dr. Michael Stingl

Project Supervisor
Robert Bauer
(TW Legal Tech)

Start Date
1st April 2026

Submission Date
1st October 2026

Erlangen, Germany | October 2026

ABSTRACT

Modern organizations increasingly face challenges in assigning the right employees to the right projects. Traditional workforce allocation methods often depend on manual decision-making processes, which may lead to inefficient resource utilization and suboptimal project outcomes. This thesis presents the Hybrid AI Framework for Employee Project Allocation (HAF-EPA), an intelligent recommendation system that combines machine learning and workforce analytics to support project staffing decisions.

The proposed framework integrates employee skills, project requirements, historical project performance, availability information, and collaboration characteristics into a unified recommendation pipeline. A Random Forest classifier is employed to predict employee suitability for project assignments based on engineered features extracted from multiple organizational datasets. The framework further incorporates a hybrid recommendation layer that combines predictive scores with business-oriented ranking criteria.

The system was trained using approximately 1.8 million employee-project assignment records and evaluated on 450,000 unseen test records using an 80:20 train-test split. Dataset imbalance was addressed through undersampling techniques, resulting in a balanced training dataset of 18,864 records. Experimental results demonstrate that the proposed model achieves an overall accuracy of approximately 88.98% while effectively identifying suitable employees for project assignments.

A web-based decision support application was developed to operationalize the framework. Project managers can upload project requirement documents in PDF format and receive ranked employee recommendations based on predicted suitability scores. The framework contributes to Human Resource Analytics by providing a scalable and data-driven approach to workforce allocation.

Keywords:
Employee Recommendation System, Workforce Allocation, Human Resource Analytics, Machine Learning, Random Forest, Data Science, Decision Support Systems.

C O N T E N T S

Introduction

Continue …………………